

Deriving forest stand information from small sample plots: An evaluation of statistical methods

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ABSTRACT

Most strategic and operational forest management decisions are taken based on stand-level information, and quantitative models of forest dynamics are key for developing sustainable management strategies. However, data on forest stands for the initialisation of such models that are representative at large spatial scales, e.g., countries or ecoregions, are often lacking. National Forest Inventories (NFIs) provide forest data from small sample plots at large spatial scales, yet deriving full stand information based on such data is challenging. Here, we evaluate seven methods of varying complexity for deriving quantitative stand descriptions based on sample data as provided by the Swiss NFI. We selected 271 extensively measured Swiss forests stands with unimodal diameter distributions, classified them as beech- vs. spruce-dominated in five development stages and randomly placed a small sized sample plot in each stand using the Swiss NFI sampling design (i.e., a circular plot of 500 m²). Seven modelling approaches were used to derive diameter distributions and species-specific stem numbers (i.e., tree species composition) from the sample data that are representative for a particular stand (*local scale*) and for stand types in general (*generalised scale*). The prediction performance of the modelling approaches was evaluated using 100 random samples per stand to calculate prediction errors. Generalised even-aged diameter distributions were best predicted by the simultaneous parameter prediction method (PPM), i.e. a combined three-step regression approach, with on average 1.3 to 2.5 times lower prediction errors compared to the simple pooling of diameter samples. However, uneven-aged diameter distributions were best predicted by pooling. At the local scale, the simultaneous PPM performed best for data from sample plots with fewer than 17 to 19 trees across all development stages. Prediction performance of the PPMs increased for structurally and spatially diverse local stands with positively skewed diameter distributions. A Random Forest approach was most suitable for predicting species composition at both the generalised and the local scale. Our study evaluates the strengths and weaknesses of methods to model stands based on data from small sample plots. We emphasise terminological pitfalls by consequently distinguishing local accuracy and generalised representativity of the stand descriptions. We demonstrate the feasibility of deriving locally accurate stands using data from small forest sample plots and evaluate the derivation of generalised stands representative at large regions. At both scales, our developments contribute to an improved initialisation of forest models and thus to a more realistic modelling of forest development under future boundary conditions.

1. Introduction

Forest policies worldwide are committed to a sustainable use of forest resources (European Commission, 2021; Katila et al., 2019). However, implementation recommendations of sustainable strategies

such as the 'New EU Forest Strategy for 2030' (European Commission, 2021) are usually formulated in general terms and remain vague when it comes to concrete implementation at the stand level, where practical forest management decisions are taken. In addition to the effects of local conditions (e.g., soil properties) on forest dynamics, forests are

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increasingly influenced by large-scale forces such as climate and forest policy changes. Therefore, foresters increasingly require decision support (Borges et al., 2014; Menzel et al., 2012) based on quantitative models of forest dynamics (e.g., Mina et al., 2017; Temperli et al., 2012; Thrippleton et al., 2021). However, accurate stand data that can be used for model initialisation are often lacking, particularly at larger spatial scales (e.g., countries, eco-regions). Stand data for model initialisation are often derived based on simplified assumptions, e.g., from pooled sample data (Huber et al., 2021), mean stand attributes directly derived from small sample plots (Kunstler et al., 2013), yield tables (Shanin et al., 2011), or spin-up model runs from bare ground with little empirical grounding (Elkin et al., 2013; Thom et al., 2018).

Ideally, stand descriptions are based on extensively measured plots that are large enough to reflect the characteristics of the entire stand. Data from Growth-And-Yield networks (Forrester et al., 2019; Seynave et al., 2018) and extensively measured natural reserves (Brang et al., 2011; Zhu et al., 2019) provide such locally accurate stand descriptions, i.e. size, species and location measurements of individual trees within spatially delimited stands at a certain time. Such detailed empirical data can be directly used to accurately initialise stand simulators at the local scale (e.g., Schmid et al., 2006; Zhu et al., 2019). Furthermore, stand data can be summarized into categories of stands with common features (e.g., by development stage or forest type), which in turn can be used to derive ‘generalised’ stand descriptions (i.e., resembling a stratum such as ‘uneven-aged beech-dominated forest’) and are representative for a set of stands in a study region, such that they may be useful for generalising scientific results (Huber et al., 2021; Mey et al., 2021). However, empirical data on extensively measured stands is usually only available locally for small regions. Thus, such data cannot normally be used for initialising forest models at larger and representative spatial scales, and they are also of limited value for generalising stand descriptions.

National Forest Inventories (NFIs) provide representative large-scale forest samples, but sample plots are small, e.g., circular plots of 500 m² in countries such as Greece, Hungary, Ireland, Switzerland and Argentina (Vidal et al., 2016), and obviously they do not represent full forest stands. For example, whether data from a single NFI sample plot of a large tree population (stand) of several hectares contains a certain species depends on the sample size, the abundance of the species within the population and the spatial aggregation of the species (Plotkin and Muller-Landau, 2002). Further, NFI sample data are usually limited by calliper thresholds as not every tree in the sampled area is measured (e.g., diameter thresholds of 12 and 36 cm for the Swiss NFI, and 4.5 and 9.5 cm for the Italian NFI, respectively; Vidal et al., 2016), which must be considered when extrapolating to the stand level (e.g., by weighting or truncation). However, a large advantage of NFIs is the high number of sample plots (typically several thousand plots) systematically distributed across large areas (e.g., a regular grid of 1.4 × 1.4 km across 41,285 km² in Switzerland). They therefore provide a representative and unbiased estimate of the sampled populations in the area covered by the sample data (Fuller, 2009; Magnussen and Renaud, 2016) and can jointly be used for model-based inference.

In this study, we used and evaluated multiple modelling approaches based on large-scale forest sample data that are targeted to 1) cope with the restrictions originating from the sampling described above, 2) improve stand descriptions at local and generalised scales, and 3) can be used to initialise forest models at either scale.

Mey et al. (2021) proposed a framework to derive spatially representative and realistic stand descriptions based on data from small sample plots that can be used for initialising forest models at large spatial scales. However, despite the supposed advantages, a detailed assessment of model accuracy and a comparison to simpler modelling approaches, such as using the sample data directly to derive stand-level information, is missing. Further, Mey et al. (2021) focused on generalised representations of stands and did not derive stand descriptions at the local scale, even though this would be feasible with the proposed

method. Such locally accurate stand descriptions derived from NFI data can be particularly useful for investigating stand-specific effects of sustainable management interventions at larger scales (e.g., countries, ecoregions) using dynamic forest models. A more detailed model initialisation may improve both the accuracy of and the stakeholders’ confidence in predictions. However, the initialisation of forest models at larger scales should consider the inherent trade-off between local-scale accuracy and generalised applicability and representativity (e.g., a region or a stratum of similar forest type) of the stand descriptions. That is, a local stand description does not represent forests at larger spatial scales, while a generalised (e.g., stratum-specific) stand description neglects local features of stands at smaller spatial scales.

The aim of the present study is to systematically evaluate a wide range of modelling approaches for deriving stand-level information using data from small sample plots on both local and generalised scales. We used extensively measured data from the Swiss Growth-And-Yield network (Forrester et al., 2019; Forrester et al., 2021) (Experimental Forest Management (EFM) network) and Natural Forest Reserves (NFR) (Brang et al., 2011; Hobi et al., 2020) in Switzerland. We addressed the following questions:

- Which methods are useful to derive local and generalised stand descriptions based on data from small sample plots that representatively cover large spatial regions, and how do they perform at the local vs. the generalised scale?
- How can these methods be used to improve the multi-scale initialisation of dynamic forest models?

2. Material and methods

2.1. Data

Our analyses were based on data from the Swiss networks of Experimental Forest Management (EFM) (Forrester et al., 2019; Forrester et al., 2021) and Natural Forest Reserves (NFR) (Brang et al., 2011; Hobi et al., 2020), covering 271 stands (EFM: 181; NFR: 90) that have been extensively measured between 1980 and 2018. The EFM network covers a wide range of silvicultural treatments (e.g., thinning, planting spacing, plentering) and all major forest types in Switzerland (Forrester et al., 2019). The NFR network was established in the late 1940s, with some stands being unmanaged or only extensively managed for centuries, while others were intensively managed until the beginning of the 20th century (Brang et al., 2011). The plots had an average size of 5,141 m² (minimum = 1,004 m², maximum = 30,141 m²). Each plot comprised georeferenced single-tree measurements of diameter at breast height (dbh) and tree species above a threshold of 8 cm (EFM) and 4 cm dbh (NFR). The stands were classified into different development stages based on the dominant diameter (d_{dom} : mean dbh of the 100 thickest trees per hectare) following the Swiss NFI (see Fischer and Traub, 2019; pole stage: $d_{dom} = 12$ –30 cm, young: $d_{dom} = 30.1$ –40 cm, medium: $d_{dom} = 40.1$ –50 cm, mature: $d_{dom} > 50$ cm, uneven-aged: 100 thickest trees spread across at least three d_{dom} classes). Most of the NFR plots were classified as uneven-aged forests (58.9%), while the EFM plots were often classified as even-aged (i.e., pole, young, medium, mature) (73.5%, see Table 1). The plots were further classified by forest types. These were defined as beech-dominated (share of beech basal area > 50%), spruce-dominated (share of spruce basal area > 50%) and ‘others’, and were used to derive forest type-specific species composition including 11 tree species groups (Table 2). We used the Swiss NFI survey design (Fischer and Traub, 2019) to draw sample data from the completely measured data of the EFM and NFR plots (Table 1; details in Section 2.4).

To guarantee that all investigated stands feature a unimodal diameter distribution, we filtered the data as follows: In a first step, all trees with a dbh below half of the minimum d_{dom} used to classify the development stage (diameter threshold, see Table 1) were omitted. This step

Table 1
Summary statistics of the stand characteristics (mean, 5% and 95% quantile in brackets) used to derive diameter distributions from the EFM and NFR plots (full data) and the data from small sample plots (circular plots of 500 m²) drawn from the EFM and NFR plots in the 100 replicates of the simulation experiment. Trees with a dbh below the 'diameter threshold' (half of the minimum d_{dom} used to classify the development stage, see Fischer and Traub, 2019) were omitted to ensure unimodality of the diameter distributions in the different development stages.

Variable	Acronym	Unit	Development stage				
			Pole	Young	Medium	Mature	Uneven-aged
Full data	Diameter threshold	cm	7.5	15	20	25	7.5
	Plot area	m ²	1,908 (1,140, 3,933)	3,440 (1,050, 10,872)	4,125 (1,131, 10,882)	5,670 (1,953, 11,808)	8,108 (2,358, 20,012)
	Number of stands (EFM; NFR)	ns	58 (52, 6)	43 (30, 13)	17 (17, 0)	52 (34, 18)	101 (48, 53)
	Number of trees per stand	nt	205 (71, 374)	181 (46, 488)	147 (24, 345)	172 (49, 437)	473 (95, 1,144)
	Dominant diameter	ddom	23.2 (15.7, 30.0)	35.5 (30.4, 39.9)	46.2 (41.2, 49.8)	60.8 (50.8, 75.7)	51.4 (40.5, 65.5)
	Quadratic mean diameter	qmd	15.8 (11.8, 20.9)	27.3 (22.8, 33.9)	39.3 (34.6, 49.8)	51.1 (39.1, 69.8)	30.6 (23.8, 39.0)
	Spread of diameters	spr	22.1 (8.83, 38.3)	29.4 (18.7, 41.3)	37.3 (20.2, 75.0)	57.3 (35.7, 92.4)	75.9 (45.4, 117.9)
	Diameter 10% quantile	dq10	9.77 (8.15, 14.3)	18.9 (15.8, 26.3)	28.2 (23.3, 38.3)	34.9 (27.5, 50.0)	10.5 (8.3, 15.5)
	Diameter 90% quantile	dq90	20.9 (14.6, 28.7)	34.7 (30.0, 40.4)	48.4 (41.8, 59.7)	65.2 (50.9, 85.9)	48.3 (34.5, 63.9)
	Number of sample plots	ns	57 (55, 58)	42 (40, 43)	16 (15, 17)	48 (46, 50)	97 (95, 100)
Sample data	Number of trees per sample plot	nt	15 (6, 23)	13 (5, 33)	14 (6, 28)	12 (4, 22)	14 (5, 27)
	Dominant diameter	ddom	22.4 (14.9, 29.7)	34.4 (26.5, 41.3)	45.8 (38.4, 53.7)	59.0 (45.2, 79.6)	49.3 (34.6, 67.3)
	Quadratic mean diameter	qmd	16.0 (11.2, 21.9)	27.4 (21.3, 34.5)	38.1 (32.3, 44.2)	50.6 (36.0, 70.1)	32.2 (20.2, 47.5)
	Spread of diameters	spr	13.7 (4.95, 23.7)	19.9 (9.35, 30.4)	22.6 (9.70, 36.7)	33.7 (14.8, 67.2)	48.5 (25.6, 80.4)
	Diameter 10% quantile	dq10	10.7 (7.90, 15.8)	19.5 (15.2, 27.1)	28.7 (20.9, 36.6)	36.3 (25.4, 56.4)	14.9 (7.90, 28.0)
	Diameter 90% quantile	dq90	20.8 (13.8, 28.3)	34.3 (26.0, 43.0)	46.6 (40.4, 55.1)	64.2 (44.8, 92.0)	48.7 (28.2, 74.8)

was necessary because many EFM plots classified in the development stages young, medium and mature showed advanced regeneration and therefore featured bimodal diameter distributions, which cannot be properly described by unimodal functions. In a second step, the modality of the diameter distributions was again visually checked with histograms and kernel density estimation (KDE; see Eq. (2) below) using Scott's rule of thumb (Scott, 1992). The stands that still featured a clear bimodal diameter distribution were excluded from the analysis.

2.2. Modelling diameter distributions

For describing the diameter (hereafter d) distribution of stands based on extensively measured data, we used the unimodal three-parameter Weibull probability density function (PDF) (Bailey and Dell, 1973) that comprises a shape (κ), scale (λ) and location parameter (α) with $d, \kappa, \lambda, \alpha > 0$:

$$f(d; \kappa, \lambda, \alpha) = \frac{\kappa}{\lambda} \left(\frac{d - \alpha}{\lambda} \right)^{\kappa-1} e^{-\left(\frac{d - \alpha}{\lambda} \right)^{\kappa}} \quad (1)$$

Additionally, we used non-parametric kernel density estimation (KDE; Parzen, 1962; Rosenblatt, 1956) to describe diameter distributions without assuming any predefined form of diameter distribution:

$$g_h(d) = \frac{1}{n} \sum_{i=1}^n k_h(d - d_i) \quad (2)$$

where d_i are the observed diameters from a distribution G with density g and k is the density of a Gaussian smoothing kernel with the bandwidth h determined by Scott's rule of thumb (Scott, 1992). The full diameter data of the EFM and NFR plots and Eqs. (1) and (2) were used to estimate diameter distributions for each development stage based on all the stands that belonged to it (generalised scale) and for individual stands (local scale). These diameter distributions were used as benchmarks for evaluating the performance of the models based on sample data only (see Section 2.2.2).

In contrast to the fully measured stands, NFI sample plots are small (500 m²) and the sampled trees usually have diameter-dependent inclusion probabilities that need to be considered to derive diameter distributions from the sample data. We used two ways to adjust the unimodal three-parameter Weibull PDF by diameter-specific calliper thresholds of the Swiss NFI: First, (multi-) truncation (Nanos and Sjöstedt de Luna, 2017):

$$f(d_{ij}; \kappa, \lambda, \alpha, T_j) = \frac{\kappa}{\lambda} \left(\frac{d_{ij} - \alpha}{\lambda} \right)^{\kappa-1} e^{-\left(\frac{d_{ij} - \alpha}{\lambda} \right)^{\kappa}} \quad (3)$$

where T_j (the truncation parameter) is the lower limit of diameter d per diameter class j ($7.5 \leq d$ less than 12, $12 \leq d$ less than 36, $d \geq 36$) defined as $T_1 = 7.5$ cm, $T_2 = 12$ cm, and $T_3 = 36$ cm. Second, (multi-) weighting (Mehtätalo et al., 2011):

$$f(d_{ij}; \kappa, \lambda, \alpha, W_j) = \frac{W_j * f(d_{ij}; \Phi)}{\int_a^\infty W_j * f(x_{ij}; \Phi) dx} \quad (4)$$

where $f(d_{ij}; \Phi)$ is the three-parameter Weibull PDF (Eq. (1)) and W_j are the weights per diameter class j , which account for the area ratios of the circular plots in the Swiss NFI: $W_1 = (4/12.62)^2$, $W_2 = (7.98/12.62)^2$, $W_3 = 1$, with 4 m, 7.98 m and 12.62 m as the plot radii.

2.2.1. Model approaches

A linear regression analysis based on the extensively measured EFM and NFR data including development stage, forest type and tree species group as explanatory variables showed that development stage was most important for determining the form and location parameter of the

Table 2
Summary statistics of the data from small sample plots (circular plots of 500 m²) of the EFM and NFR plots used as explanatory variables for modelling tree species composition in the 100 replicates of the simulation experiment. The relative tree diameter per sample plot was calculated by $(d - \min(d))/(\max(d) - \min(d))$.

Variable	Category	Unit	Quantiles				
			0%	5%	50%	95%	100%
Tree species group	Beech (<i>Fagus sylvatica</i>)	Number of trees	301	312	332	360	401
	Douglas-fir (<i>Pseudotsuga menziesii</i>)		30	33	39	61	65
	Larch (<i>Larix decidua</i> , <i>L. kaempferi</i>)		352	368	394	434	448
	Maple (<i>Acer pseudoplatanus</i> , <i>A. campestre</i> , <i>A. platanoides</i> , <i>A. opalus</i>)		186	200	219	239	258
	Oak (<i>Quercus robur</i> , <i>Q. pubescens</i> , <i>Q. petraea</i> , <i>Q. rubra</i>)		377	391	417	436	444
	Other broadleaves (<i>Carpinus betulus</i> , <i>Prunus</i> spp., <i>Tilia</i> spp., <i>Salix</i> spp., etc.)		118	125	145	167	181
	Other conifers (<i>Taxus baccata</i> , <i>Cedrus</i> spp., <i>Pinus ponderosa</i> , etc.)		55	57	64	72	77
	Pine (<i>Pinus sylvestris</i> , <i>P. mugo</i> , <i>P. nigra</i> , <i>P. strobus</i>)		271	284	321	347	362
	Silver fir (<i>Abies alba</i>)		212	221	255	287	296
	Spruce (<i>Picea abies</i>)		898	919	960	1022	1043
	Swiss pine (<i>Pinus cembra</i>)		125	132	151	165	188
	Beech-dominated		15	18	21	26	27
	Spruce-dominated		50	52	59	66	69
	Other		154	162	170	176	179
Forest type		Number of plots	338	427	885	1937	2186
Elevation		m a.s.l.	0.22	2.11	14.89	37.69	40.44
Slope		degrees	0.00	0.00	0.45	1.00	1.00
Relative tree diameter per plot		–					

diameter distributions. The forest type and species group had no significant effect. Thus, only the development stage was used to categorise the diameter distribution of the stands.

2.2.1.1. Pooling. The most straightforward approach to derive development stage-specific diameter distributions from sample data is pooling. Thereby the weighted (by the representation factors that account for the calliper thresholds in the Swiss NFI and the respective sampled area, see Section 2.4) diameters of sample plots classified as the same development stage were pooled. Afterwards KDE (Eq. (2)) was applied to derive the pooled development stage-specific diameter distributions. Pooling neglects characteristics of local diameter distributions (e.g., high abundance of smaller trees, absence of larger trees). Thus, compared to the diameter distributions derived from individual stands, particularly small and large trees may be under- or overrepresented in the pooled data, and the approach is therefore not suitable for describing the diameter distributions of local stands.

2.2.1.2. Direct fit. Local diameter distributions can be obtained by fitting density functions directly to data from individual sample plots. We fitted (by Maximum Likelihood Estimation, MLE) the truncated and the weighted Weibull function (Eqs. (3) and (4)) directly to the diameter sample of each plot to obtain local diameter distributions. However, each direct fit solely builds on the diameters of the respective sample plots, i.e. an excerpt of the diameter data of the complete stand, and therefore only partly reflects the diameter distribution of a complete stand.

2.2.1.3. Traditional PPM. In contrast to the pooling and the direct fit approach, the parameter prediction method (PPM) accounts for both general and local stand characteristics as it links diameter distributions through regression to explanatory stand variables. The traditional PPM (Clutter and Bennett, 1965; Hyink and Moser, 1983) comprises three steps (cf. Mehtätalo and Lappi, 2020): 1) Selecting a density function that is useful to describe the diameter data, 2) estimating the parameters of the density function for the diameter distribution of each sample plot (by MLE) and 3) using the estimated parameters as response and stand characteristics as explanatory variables in linear regression models.

The regression approach of the traditional PPM enables evaluating model quality (e.g., by AIC, R²) and may reduce errors of diameter predictions at local and generalised scales. We applied the traditional PPM based on the truncated (equation 8 in Nanos and Sjöstedt de Luna, 2017) and the weighted Weibull function (Mehtätalo et al., 2011) to the sample data, respectively. Development stage-specific (i.e., generalised) diameter distributions were obtained using median stand characteristics of the sample plot data classified in the different development stages as explanatory variables in the PPM. Local diameter distributions were obtained using the stand characteristics of data from each individual sample plot.

2.2.1.4. Simultaneous PPM. Cao (2004) combined the steps of the traditional PPM described above in one closed formula that allows for direct model and parameter inference (see also Mey et al., 2021). Thereby the Weibull parameters to describe the diameter distribution using data from a set of sample plots are estimated jointly by parameter-specific linear regression models and MLE (Eq. (5) by Mey et al., 2021). This approach resembles a generalised linear model (GLM) with the response variable being Weibull distributed (Mehtätalo and Lappi, 2020) and may reduce errors of diameter predictions in comparison to the traditional PPM. We used the unimodal diameter model by Mey et al. (2021), hereafter the simultaneous PPM, based on the truncated and weighted Weibull function, respectively, and predicted local and development stage-specific (i.e., generalised) diameter distributions as with the traditional PPM.

2.2.2. Evaluation of the accuracy of the predicted diameter distributions

As a benchmark for evaluating the prediction performance of the different approaches, we determined diameter distributions for each development stage category (reference for development stage-specific diameter distributions) and for each stand ('true' local diameter distributions) using the completely measured diameter data of the EFM and NFR plots.

2.2.2.1 Reference for development stage-specific diameter distributions. We used the traditional PPM approach (cf. Section 2.2.1) based on the completely measured diameter data of the EFM and NFR plots to predict references for development stage-specific diameter distributions. The traditional PPM was chosen because test runs showed a very high explanatory power of this approach (adjusted R^2 for shape > 0.85 and for scale > 0.98). In a first step, we estimated the shape and scale parameters of the three-parameter Weibull PDF (Eq. (1) for the full diameter data by maximum likelihood (MLE), while the fixed diameter thresholds (see Section 2.1 and Table 1) were directly used as location parameter. Subsequently, the estimated parameters were regressed to explanatory stand variables derived from the completely measured EFM and NFR data. The best regression models for shape and scale were selected by backward selection (*step* function in R (R Core Team, 2021)) using the Bayesian Information Criterion (BIC) (see details on model formulation in Supporting Information S1). The two regression models were then used to derive the 'reference' diameter distributions of each development stage (pole, young, medium, mature and uneven-aged) using median values of the stand characteristics of the EFM and NFR data classified in the five development stages.

2.2.2.2. True local diameter distributions. We used kernel density estimation (eq. (2) to determine the 'true' local diameter distributions based on the full diameter data of each EFM and NFR plot. To compare non-parametric (KDE) and parametric predictions (Weibull function) in general, we also estimated (via MLE) the three-parameter Weibull function based on the full diameter data of each EFM and NFR plot.

2.2.2.3. Error index. The deviation between the 'reference'/'true' and the predicted diameter distributions obtained with the different methods (Section 2.2.1) was evaluated using the Error index (EI) proposed by Reynolds et al. (1988). EI quantifies the discrepancy between two PDFs with values between 0 (exactly equal) and 2 (no overlap). Hence, a fractional EI can be calculated by dividing EI by 2. The sum in the formula of the EI by Reynolds was replaced by the integral to consider continuous diameter distributions; the formula for EI is thus given by:

$$EI = \int_{\alpha}^{\infty} |f(x) - \hat{f}(x)| dx \quad (5)$$

where the integral starts at the location parameter (α), with $f(x)$ being the 'reference'/'true' and $\hat{f}(x)$ the estimated three-parameter Weibull PDF.

2.3. Modelling tree species composition

We modelled tree species composition based on the stem shares of 11 main tree species groups in Switzerland. The 'true' local species composition was derived directly from the full species data of the individual EFM and NFR plots by calculating the relative shares of each species group per plot:

$$p_i = \frac{n_i}{\sum_{i=1}^N n_i} \quad (6)$$

where n_i is number of stems of the species group i , N is the total number of stems per plot, and p_i is the relative share of the individual species with $\sum p_i = 1$.

The 'reference' forest type-specific species composition (spruce-dominated; beech-dominated; other) was determined by merging the full species data from the EFM and NFR plots comprising these forest types and calculating the relative shares of the individual species per forest type:

$$p_{ij} = \frac{\sum_{j=1}^{N_j} n_{ij}}{\sum_{j=1}^{N_j} \sum_{i=1}^N n_{ij}} \quad (7)$$

where n_{ij} is the stem number of species i of forest type j , N_j is the number of sample plots classified as forest type j , and p_{ij} is the forest type-specific relative share of the individual species.

2.3.1. Model approaches

To predict tree species composition based on data from small sample plots, three approaches were considered. Each approach weights individual trees using the representation factors that account for the calliper thresholds and the sampled area of the three sub-plots in the Swiss NFI (cf. Section 2.4): 1) Observed weighted stem shares by species, 2) multinomial logistic regression, and 3) a Random Forest. Random Forest is a decision-tree based ensemble method (Breiman, 2001) that can be used for regression and classification problems. It does not require any data transformation, is robust against outliers, can handle imbalanced data and automatically performs a cross-validation (out-of-bag).

Methods 2) and 3) were applied using the representation factors as case weights in model fitting. Since we were interested in in-sample predictions only, i.e. predicting for each sample plot the underlying species composition of the stand, no splitting in training and test data or cross-validation was performed. We used a set of 20 explanatory variables (see S3) to predict the occurrence probabilities of 11 tree species groups in the multinomial logistic regression following Mey et al. (2021) as well as in the Random Forest approach.

For the multinomial logistic regression, backward model selection was performed using the BIC. The final multinomial logistic regression comprised 10 explanatory variables (cf. S3).

The Random Forest (RF) approach (for classification) as implemented in the *ranger* package (Malley et al., 2012) was used for consistent estimations of probability and case weights. The best predictive RF was derived based on model selection with the out-of-bag (OOB) error rate using the *VSURF* package (Genuer et al., 2015). The final RF comprised the four explanatory variables relative diameter, slope, elevation and plot ID, with plot ID accounting for stand-specific effects (e.g., reproduction advantages of the dominant species) but also limiting predictions of the model to the investigated stands (in-sample). Using the relative diameter as predictor in the species models, the predictions of diameter distributions and species composition were linked (cf. Mey et al., 2021).

2.3.2. Evaluation of the accuracy of predicted species composition

Weighted observed species shares, multinomial logistic regression and RF were used to estimate the local species composition of each stand, i.e. the relative stem shares by species. To obtain the predicted forest type-specific species composition of spruce-dominated, beech-dominated and other forests, we merged the local relative species shares predicted for the sample plots classified as the same forest type (cf. Section 2.1).

The accuracy of the model predictions was evaluated based on the over- and underestimation (positive and negative probability error rates, respectively) of the relative stem shares of the 11 tree species groups in the stands, i.e. the positive or negative deviation of the predicted occurrence probabilities of the species relative to the observed occurrence probabilities. Thereby, we distinguished between local and forest type-specific species composition, as described above.

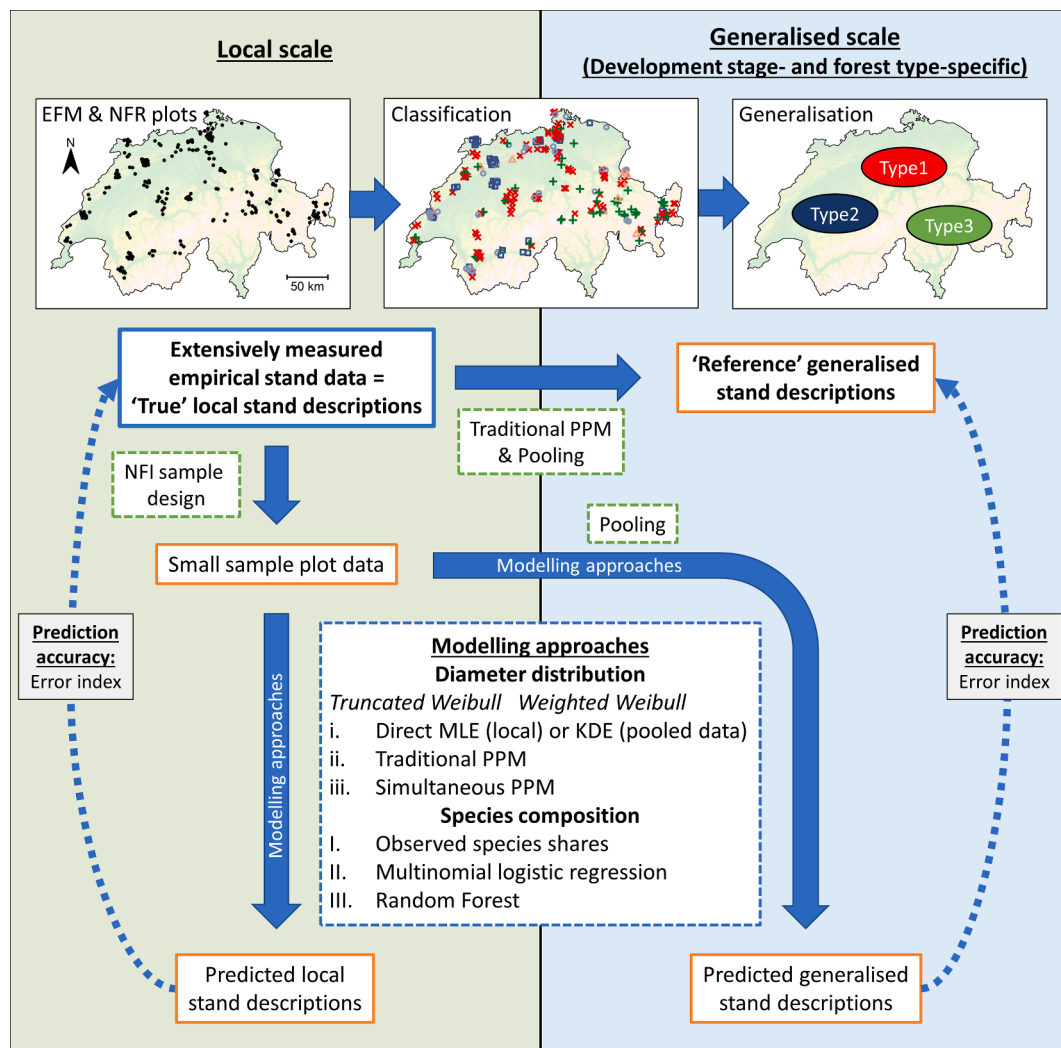


Fig. 1. Evaluation scheme for deriving stand-level information (diameter distribution and tree species composition) based on data from small sample plots at local (left) and generalised scales (right). Four approaches (direct maximum likelihood estimation (MLE), kernel density estimation (KDE), traditional parameter prediction method (PPM) and simultaneous PPM based on the truncated and weighted Weibull PDF) were used to predict stand diameter distributions and three approaches (observed species shares, multinomial logistic regression, and Random Forest) were used to predict stand species composition from the samples. In the maps of Switzerland, the location of the 271 EFM and NFR plots are jittered to reduce overlapping of clumped plots.

2.4. Simulation experiment

To evaluate the prediction accuracy of the different modelling approaches for deriving stand descriptions from sample data, a simulation experiment was conducted. Thereby we distinguished model performance on local and generalised (stratum) scale (in the case of the diameter models: development stage-specific; in the case of the species model: forest type-specific), see Fig. 1. The steps of the random sampling procedure, model building on the sample set and subsequent model evaluation described below were reproduced 100 times. Thus, the performance of the different model approaches was evaluated on 100 different sample sets drawn from the 271 EFM and NFR plots.

The revised data set of the EFM and NFR plots (see Section 2.1) was used to draw sample data from each plot according to the circular Swiss NFI sample design (see Fischer and Traub, 2019; Mey et al., 2021). We distinguished the three concentric sample circles 500 m², 200 m² and 50.27 m² in which trees were sampled with diameter-specific calliper thresholds ($d \geq 32$ cm, $d \geq 12$ cm and $d \geq 7.5$ cm). The location of the sample plot was chosen randomly, provided that the largest sample circle (500 m²) was entirely located within the EFM or NFR plot area. Sample plots with less than four trees were excluded from the analysis in

order to use always at least the number of parameters + 1 trees for the estimation of the three-parameter Weibull function (Eq. (1)).

Subsequently, the traditional and simultaneous PPM were applied based on the truncated and weighted unimodal Weibull function to derive development stage-specific and local diameter distributions from the sample data. In addition, local diameter distributions were determined by fitting (by MLE) the truncated and weighted Weibull PDF directly to the sample data, while KDE was used on the (weighted) sample data pooled by development stage. The diameter thresholds (see Section 2.1 and Table 1) were used as the location parameter (minimal possible diameter) in the three-parameter Weibull functions (Eqs. (1), (3) and (4)) of the different modelling approaches. The shape parameter was log-transformed to obtain approximately normally distributed residuals, which was not necessary for the scale parameter. The two best explanatory variables for the shape and scale parameter of the truncated and weighted PPM approaches, respectively, were selected using the exhaustive selection algorithm of the *regsubsets* function (minimising the residual sum of squares) in the *leaps* package (Lumley, 2020). We limited the models to two explanatory variables per parameter since each additional variable increases the complexity of the simultaneous PPM and thus hampers parameter optimisation. We maximised the log-

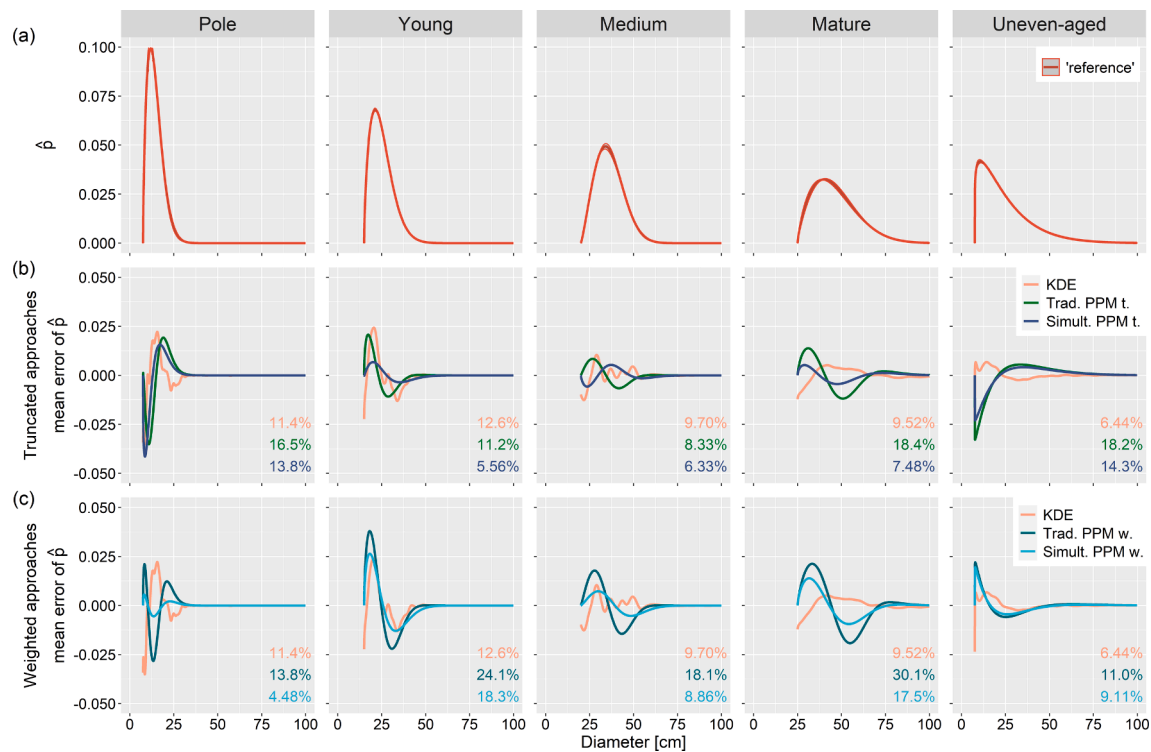


Fig. 2. Mean ‘reference’ for development stage-specific (generalised) diameter distributions (a) and mean error to the modelling approaches derived from 100 simulation runs and thus sample sets. That is (b) kernel density estimation (KDE), traditional parameter prediction method (PPM) and simultaneous PPM based on the truncated Weibull PDF, and (c) KDE and PPMs based on the weighted Weibull PDF. The mean percentual Error index per method and development stage is given in the lower right corner of each plot.

Table 3

Mean coefficients ($\pm$ mean standard errors) of the two best explanatory variables per parameter (shape (κ), scale (λ)) using the truncated and weighted Weibull function in the traditional and simultaneous parameter prediction method (PPM) in 100 simulation runs.

Explanatory variable	Parameter estimates							
	Truncated Weibull				Weighted Weibull			
	Traditional PPM		Simultaneous PPM		Traditional PPM		Simultaneous PPM	
	$\log(\kappa)$	λ	$\log(\kappa)$	λ	$\log(\kappa)$	λ	$\log(\kappa)$	λ
Intercept	0.536 ± 0.024	14.68 ± 0.390	0.399 ± 0.017	14.58 ± 0.216	0.726 ± 0.022	19.18 ± 0.363	0.563 ± 0.017	18.90 ± 0.230
Spread of diameters	-0.373 ± 0.025	-2.80 ± 0.422	-0.216 ± 0.014	-1.14 ± 0.235	-0.434 ± 0.023	0.023	-0.292 ± 0.017	0.017
Quadratic mean diameter		9.30 ± 0.416		7.80 ± 0.202	0.320 ± 0.024	12.67 ± 0.693	0.239 ± 0.018	12.63 ± 0.417
Diameter 10% quantile	0.307 ± 0.025		0.232 ± 0.017			0.662 ± 0.662		-4.62 ± 0.441

likelihood of the simultaneous PPM with the numerical algorithm ‘L-BFGS-B’ in the R function *optim* (R Core Team, 2021).

3. Results

3.1. Development stage-specific diameter distributions

The linear regression models used to derive the ‘reference’ development stage-specific diameter distributions (Fig. 2a) explained the Weibull parameters very well (mean adjusted $R^2 \pm$ SD for shape = 0.87 ± 0.0102 and for scale = 0.99 ± 0.0007). The development stage-specific diameter distributions showed reasonable diameter ranges with larger diameters and broader distributions in later development stages, and a heavily positively skewed distribution (almost inverse J-shaped) under uneven-aged conditions.

The quadratic mean diameter, the spread of diameters, and the 10% quantile of the diameter (in order of the mean effect size) were selected

as the best explanatory variables in the traditional and simultaneous PPM (Table 3), whereby the PPMs based on the truncated Weibull function led to generally lower effect sizes (coefficient values) than the PPMs based on the weighted Weibull function. Across all development stages, the simultaneous PPM approach by Mey et al. (2021) produced lower mean Error indices (EI) compared to the traditional approach (Fig. 2b and c). Differences in EI between traditional and simultaneous PPM were highest in mature stands (11.8%) and lowest in uneven-aged stands (2.9%). For even-aged stands, the simultaneous PPM based on either the truncated (Fig. 2b) or the weighted Weibull (Fig. 2c) showed better prediction performances than the pooling approach with subsequent kernel density estimation (KDE). The weighted Weibull function performed better in the development stages pole and uneven-aged, while using the truncated Weibull led to lower Error indices in the development stages young, medium and mature. Uneven-aged stands were best predicted by pooling and subsequent KDE (mean EI = 6.4%).

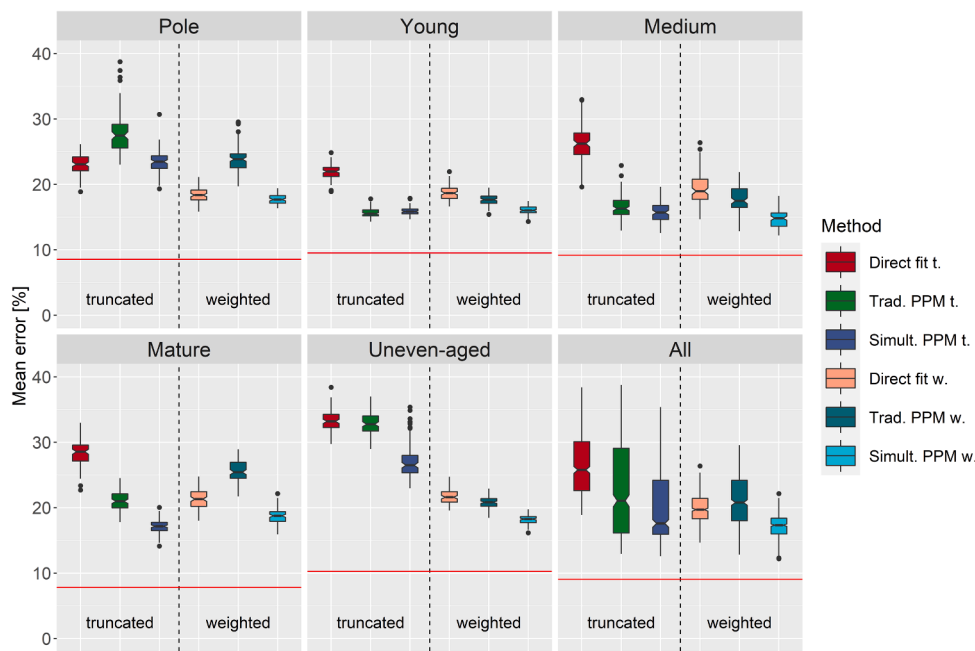


Fig. 3. Mean Error index (in %) of local diameter distributions derived from the different methods using the truncated Weibull PDF (left) and the weighted Weibull PDF (right) averaged over the stands within the individual (pole, young, medium, mature, uneven-aged) and all development stages in 100 simulation runs. The red horizontal lines indicate the mean error between the non-parametric kernel density estimation (KDE) and the three-parameter Weibull fitted to the full data of the EFM and NFR plots as a benchmark.

3.2. Local diameter distributions

Compared to the non-parametric KDE (representing the ‘true’ local diameter distributions), the parametric Weibull approaches increased the mean error for the prediction of the local diameter distributions by 8.3 to 10.6 percentage points (red lines vs. zero lines in Fig. 3). Simultaneous PPM resulted in predictions with 4.1 to 65.5% lower percentage mean errors as compared to the direct fits (except for the pole stage truncated Weibull), with the simultaneous PPM generally producing lower Error indices (Fig. 3) and lower errors of the shape and scale

parameter (Fig. S1) than the traditional PPM. The use of the weighted Weibull instead of the truncated Weibull function led to lower Error indices across the five development stages with the most methods and in most development stages, especially for uneven-aged stands (49.4 to 57.7% lower EI). However, in the PPMs, the truncated Weibull reduced Error indices for young (by 1.1 percentage points) and mature stands (by 3.1 percentage points) compared to the weighted Weibull function.

An explorative analysis on the influence of stand characteristics of the full data on the prediction performances (Error indices) of the direct fits and PPM approaches revealed the following effects using the

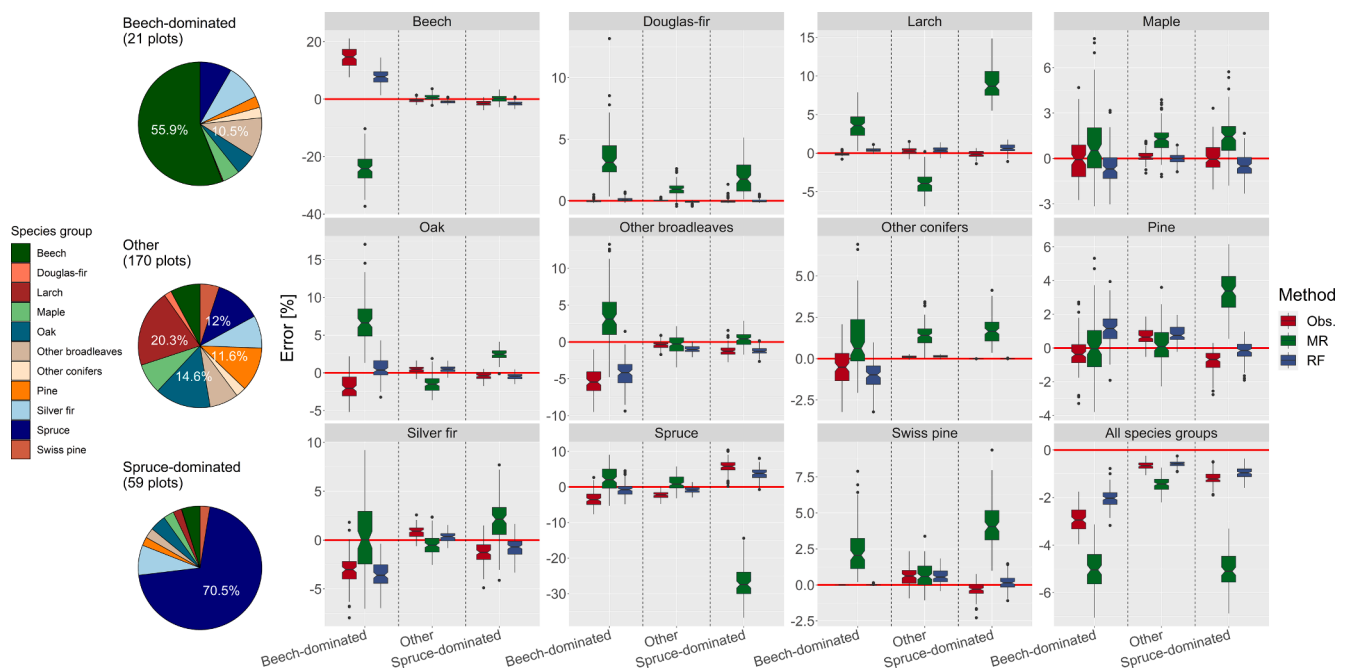


Fig. 4. Percentage prediction errors of the occurrence of the individual species groups in beech-dominated, spruce-dominated and other forests using the species models (Obs. = weighted observed relative species shares, MR = multinomial logistic regression, RF = Random Forest) in 100 sample sets. The ‘true’ forest type-specific species composition (indicated by the red reference lines in the boxplots) with the number of sample plots classified as the respective forest type are shown in the pie plots on the left.

truncated Weibull function: (a) While the error indices of the direct fits generally increased with structural and spatial diversity of the EFM and NFR plots, the error indices of the simultaneous PPM decreased for uneven-aged stands (Figure S2 and S3), (b) the area sizes of the EFM and NFR plots only marginally affected the Error indices (Fig. S4), (c) negatively skewed diameter distributions were generally better predicted by the direct fits, while positively skewed diameter distributions were mostly better predicted by the simultaneous PPM (Fig. S5). These effects were also observed for the methods that used the weighted Weibull function, but effect sizes were considerably lower (cf. Fig. S2–S5). On average, 2.9% of the data from individual sample plots were better described by a bimodal than a unimodal Weibull function (lower AIC, see Mey et al., 2021), even though all of the samples were drawn from unimodal stands (cf. Section 2.1). In the majority of cases (88.1%), these bimodal samples were drawn from uneven-aged stands. Compared to the simple direct fits to the data from individual sample plots, the simultaneous PPM improved local diameter predictions based on sample plots with less than 19 trees using the truncated Weibull function and with less than 17 trees using the weighted Weibull function (Fig. S6).

3.3. Forest type-specific tree species composition

The classification of the entire EFM and NFR data in forest types led to the ‘reference’ species composition with a relative basal area share of 70.5% for spruce in spruce-dominated stands, 55.9% for beech in beech-dominated stands and ‘other’ stands dominated by larch, oak, spruce, pine and silver fir (Fig. 4). The tree species composition of the three forest types was best predicted (lowest error rates) by the weighted Random Forest approach (mean out-of-bag error rate = 19.8%), while predictions of the multinomial logistic regression were generally least accurate. The multinomial logistic regression especially underestimated (negative error

rate) the occurrence of beech in beech-dominated stands and the occurrence of spruce in spruce-dominated stands. In contrast, the direct use of the weighted observations overestimated (positive error rate) the occurrence of beech in beech-dominated stands and of spruce in spruce-dominated stands. However, individual species were best predicted by different model approaches. For example, the multinomial logistic regression best predicted the occurrence of beech and ‘other broadleaves’ in spruce-dominated stands, while the weighted observations best predicted the occurrence of maple in beech- and spruce-dominated stands. Error rates of the different approaches typically decreased with the number of sample plots being used for the forest type-specific species composition (highest errors for beech-dominated stands based on 21 plots; lowest errors for ‘other’ stands based on 170 plots).

3.4. Local tree species composition

The Random Forest approach showed the best prediction performance regarding species composition at the local scale (mean overestimation = 8.0%, mean underestimation = 7.0%), with over- and underestimation of species occurrence fairly balanced across all species groups (Fig. 5). In contrast, the multinomial logistic regression usually underestimated species occurrence (mean underestimation = 23.2%), and led to on average 16.5 percentage points (3.4 times) higher positive errors compared to the two other methods. Especially rather rare species groups such as ‘other conifers’, larch, Douglas-fir and oak were underestimated by the multinomial logistic regression. The direct use of the weighted sample data generally led to the lowest underestimations (mean underestimation = 6.5%), but also to the highest overestimations of species occurrence (mean overestimation = 11.4%).

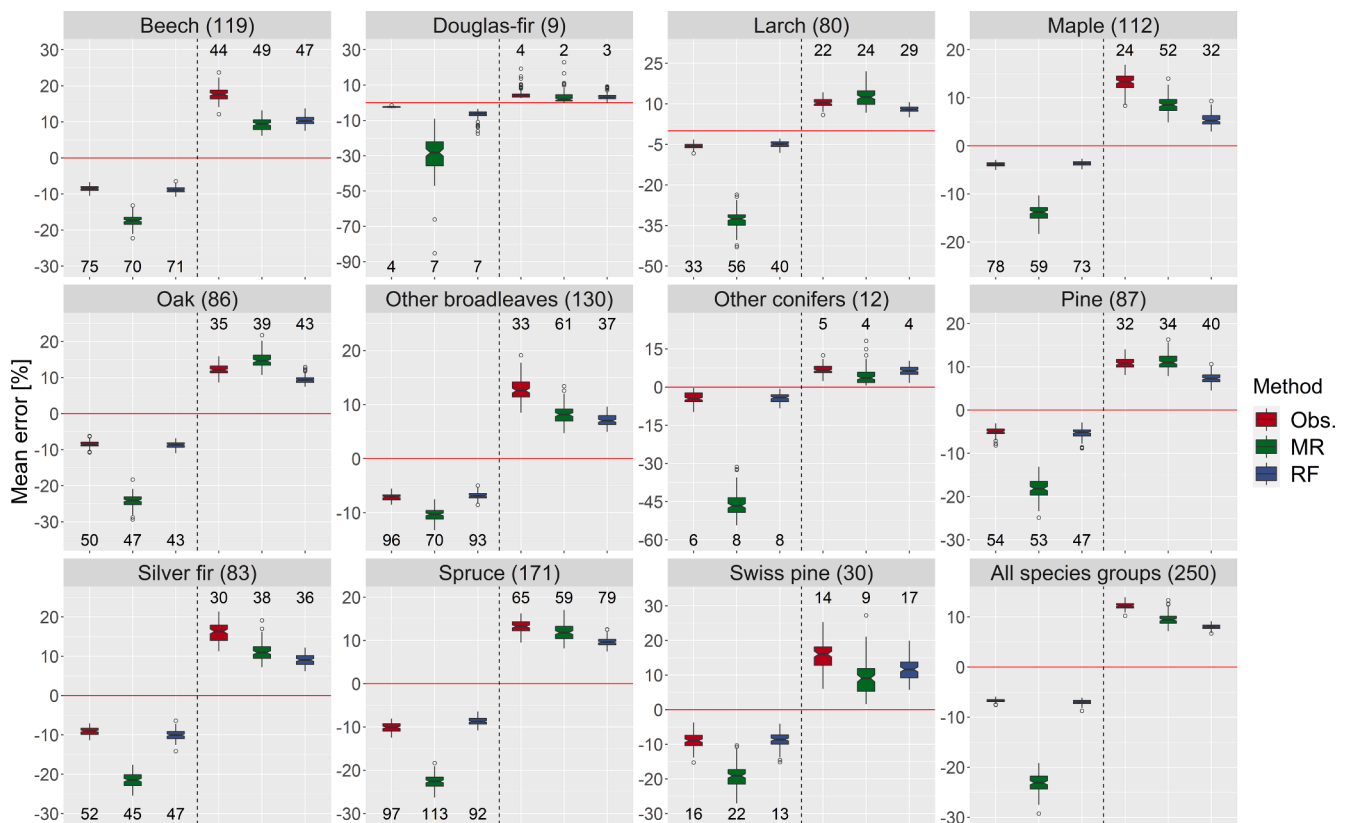


Fig. 5. Mean over- (positive percentage errors, right) and underestimation (negative percentage errors, left) of the occurrence of the local tree species groups using the weighted observed relative species shares (Obs.) and predicted with the species models (MR = multinomial logistic regression, RF = Random Forest) averaged over the individual species in 100 simulation runs. The average number of plots where the individual species was over- or underestimated is shown above or below the respective boxplot. The average total number of plots where the individual species were observed in the full EFM and NFR data is highlighted in brackets.

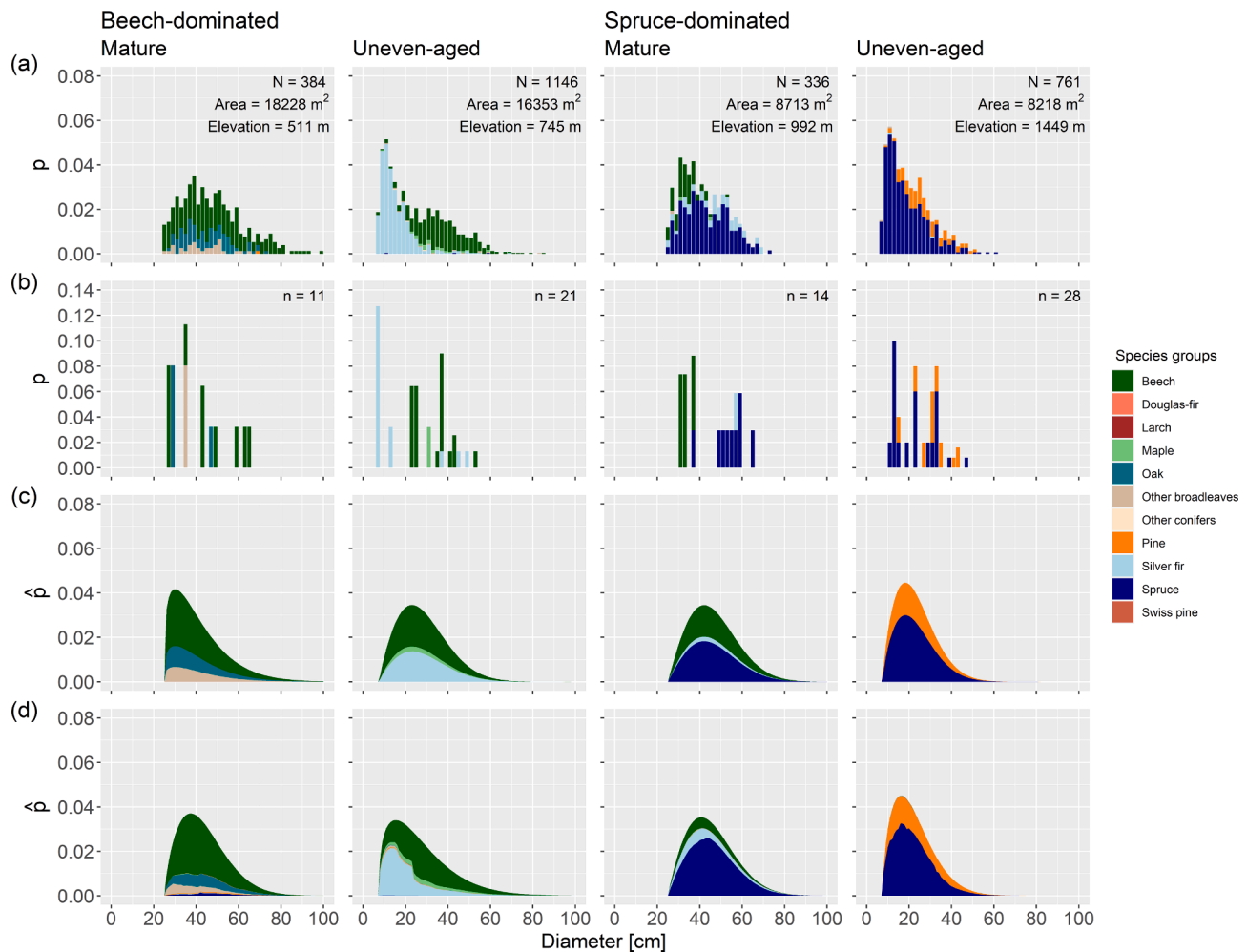


Fig. 6. Example of local stand descriptions of larger beech-dominated and smaller spruce-dominated stands in the development stages mature (left) and uneven-aged (right) using (a) the full set of the observed data (N = number of trees per hectare in the stand), (b) weighted NFI sample data (one circular sample plot of 500 m², n = number of trees per sample), (c) direct fits of the Weibull functions to the sample diameters and (d) the prediction of the simultaneous PPM and the Random Forest approach based on the sample data, where p is the observed diameter and species occurrence probability in rows (a) and (b) and $\hat{p}$ is the predicted diameter and species occurrence probability in rows (c) and (d).

3.5. Local stand descriptions: Combining diameter distribution and species composition

We identified the simultaneous PPM and the Random Forest as the best approaches to predict local stands (cf. Section 3.2 and 3.4). In contrast, (weighted) data from small sample plots (Fig. 6b) did not accurately reflect the diameter distribution and species composition of the stands from which they were drawn, especially when the stands were structured and species-rich (Fig. 6a). Usually, in the sample data entire diameter classes were not observed (i.e., tree count zero), which led to a general overrepresentation of the (weighted) sampled diameter classes in the stands. Fitting weighted Weibull functions directly to the sample diameters (Fig. 6c) improved the representation of diameter distributions of the stands, but at the same time often resulted in an overestimation (e.g., large trees in the uneven-aged beech stand) and/or underestimation of the occurrence of trees (e.g., larger trees in the mature beech stand).

A constant species composition over the entire diameter range (derived from the weighted sample data) usually inaccurately reflected stand species composition (Fig. 6c). For example, the share of beech was generally overestimated in the mature spruce-dominated stand, and the share of silver fir was underestimated in smaller diameter classes. In larger diameter classes of the uneven-aged beech-dominated stand, the share of beech was overestimated. In contrast, the combination of

simultaneous PPM and Random Forest (Fig. 6d) yielded the most reasonable diameter and species predictions over the entire diameter range including diameter-related species occurrence (e.g., higher occurrence probability of silver fir in lower diameter classes of the uneven-aged beech-dominated stand). However, the prediction of direct fit and simultaneous PPM for the uneven-aged spruce-dominated stand was similar (due to a relatively high sample size (28) in relation to stand area size), and the occurrence probability of large diameters was underestimated in the beech-dominated stands.

4. Discussion

In this study, we extended and evaluated diameter distribution and species composition models to derive local and generalised stand data from large-scale samples. Below, we discuss the modelling of generalised and local stand descriptions in the study region, and elucidate the general strengths and weaknesses, as well as the application potential of the approaches. Distinguishing local and generalised stand descriptions allows for a more targeted and improved initialisation of dynamic forest models in relation to a) the type of model, b) the available empirical data, and c) the research questions addressed by model applications.

4.1. General representativity

Generalised stand descriptions are aimed to conceptualise knowledge and advise forest management at a regional level. In forestry, generalisations of forest stand data have a long tradition and can be achieved either deductively (e.g., yield tables based on long-term experimental plots) or inductively (e.g., stratification of a large number of NFI sample plots) (Pretzsch, 2009). In this study, we followed an inductive approach and classified the NFI sample plots based on development stage and the basal area share of the dominant tree species. The classified sample data were then used to derive generalised stand descriptions (by pooling or model-based) which are representative for an entire category of stands. Other (more sophisticated) classification schemes such as the European Forest Types by Giannetti et al. (2018) can be used, but they bear the risk of leaving categories unoccupied or sample plots unclassifiable due to a limited amount of data. Generalised diameter distributions (at regional stratum level) may be useful for analysing large-scale changes of tree populations (Magnussen and Renaud, 2016). Using such representative stands to initialise dynamic forest models enables the generalisation of scientific findings, which is particularly needed for advising regional and national forest management in the face of climate change (Reyer et al., 2015). For example, representative stand types were used in simulation studies by Thrip-leton et al. (2023) to investigate the mitigation potential of different harvest intensities on disturbances predisposition for the region of Davos (Canton Grisons, Switzerland) and by Mey et al. (2022) to evaluate adaptive forest management practices to maintain regional ecosystem service provision under climate warming in Switzerland.

The diameter distributions of uneven-aged stands were best predicted by pooling diameter samples (and subsequent KDE), while the diameter distributions of even-aged stands were better predicted by the simultaneous PPM approach. This can be explained by (a) high structural diversity of uneven-aged stands with broad diameter distributions and higher probabilities of smaller trees, and thus high variability of the sample diameter distributions drawn from these stands, (b) rather narrow diameter distributions of even-aged stands featuring distinct unimodal distributions, and thus low variability of diameter samples drawn from these stands, and (c) most of the sample plots were classified as uneven-aged (on average 97), while a considerably lower number of sample plots came from even-aged stands, so that the predictive accuracy of the pooling approach was lower for such stands. The lower performance of PPM in the uneven-aged stands can be explained by the diverse forms of diameter distributions, including non-unimodal and in some cases even bimodal ones, which limited the use of the unimodal Weibull function to describe generalised uneven-aged stands.

Uneven-aged and old-growth stands typically feature inverse J-shaped or rotated-sigmoid diameter distributions (Westphal et al., 2006; Zhang et al., 2001) covering a large range of diameter classes. Rubin et al. (2006) suggested that the minimum sampling area to detect the inverse J-shaped (negative exponential) diameter distribution of old-growth stands should be one hectare. However, in our study the area of >50% of the uneven-aged stands was smaller, such that potential inverse J-shaped distributions have not been captured by our data. Past management also in NFR plots may have contributed to the development of diverse stand structures, i.e. stands with varying diameter ranges (Brang et al., 2011; Forrester et al., 2019). We thus conclude that (1) our reference diameter distribution was (heavily) positively skewed rather than inverse J-shaped, and (2) due to the diverse forms of the diameter distribution of the stands (and sample data), the PPMs either shifted the generalised diameter distribution to the left (using the truncated Weibull) or shifted to the right (using the weighted Weibull) compared to the reference. In contrast, the diameter distributions of the even-aged stands were rather homogeneously unimodal, as described e.g., by Mohler et al. (1978), and thus captured well by the simultaneous PPM. Hence, the performance of the simultaneous PPM highly depends on the PDF used (here truncated and weighted Weibull) and the accuracy of the method used to categorise the sample data.

The Random Forest provided the best predictions of forest type-specific species composition, but the difference to the other modelling approaches decreased with increasing number of sample plots. This can be explained by a combination of reasons, ordered by probable importance: (a) the advantage of a model-based assessment using all sample plots jointly (Random Forest) compared to the direct use of the observations classified as the respective forest type (especially less accurate for the beech-dominated stand with a low number of sample plots), (b) rather nonlinear behaviour of the model predictors for species composition, and (c) the underestimation of dominant tree species in the generalised stands using a logit model. Thus, the Random Forest may be especially useful to improve predictions of rather rare forest types, provided that the number of sample plots used for model fitting is high.

4.2. Local accuracy

Estimating stand-level characteristics such as mean diameter, stem numbers and growing stock from design-based sample data (e.g., NFIs) is usually unbiased (Lehtonen and Veijanen, 2009). Yet, estimating more complex quantities such as the parameter of a stand's diameter distribution may be biased (García, 2006). Forest tree populations tend to feature much more structural variability at the stand level than at the sample plot level. However, it is often assumed that diameter distributions and other size distributions obtained from single or few small forest sample plots (a few hundred m²) can be used as direct proxies for the local stand level (Borders et al., 2008; Podlaski and Roesch, 2014; Xu et al., 2011). This assumption has been shown to possibly be valid for small, spatially homogeneous (e.g., even-aged) stands, but it is difficult to justify if stand area (in relation to the sample size) along with the heterogeneity of the spatial tree pattern increases (Bankston et al., 2021; Borders et al., 2008; Gove, 2003). Our methodology improves local stand descriptions using large-scale sample data that are obtained using a design such as that of the Swiss NFI.

Our results showed that the weighted Weibull function performed considerably better for pole and uneven-aged stands than the truncated Weibull function, and the performance of both approaches decreased with increasing spatial diversity of the stands (cf. Figure S3). In an extensive simulation study, Nanos and Sjöstedt de Luna (2017) found that the performance of the truncated Weibull was comparable to the weighted Weibull function for a homogeneous spatial tree pattern, and even better for plots with an inhomogeneous tree pattern. However, they evaluated both functions solely at the sample plot level and only considered data from sample plots with > 10 trees for Error indices of shape, scale and QMD. In contrast, we used the Error index by Reynolds et al. (1988) that considers the entire diameter distribution at the stand level and calculated the index based on data from sample plots with ≥ 4 trees. Further, the discrepancy between both results may be explained by a lack of pronounced inhomogeneous tree patterns in the EFM and NFR data due to past management (Carey, 2003; Carrer et al., 2018). However, we did not further investigate the gradient of spatial patterns covered by the data, nor other measures of spatial diversity such as Ripley's K and L functions for spatial point patterns (Ripley, 1977). Thus, our results suggest that the weighted (Mehtätalo et al., 2011) rather than the truncated Weibull function is preferable for extrapolations of sample diameter distributions to the stand level. How this is related to the spatial tree pattern of stands should be investigated further.

A remarkable result was that in contrast to the generalised scale, the simultaneous PPM considerably improved the prediction of local diameter distributions in uneven-aged stands. The generally high structural diversity of uneven-aged (old-growth) stands (Franklin and Van Pelt, 2004) also applied to our study. The data from individual sample plots drawn from these stands therefore contained greater variance regarding the underlying diameter distribution than samples from even-aged stands. This variability of the diameter distributions of the individual sample plots allowed the simultaneous PPM to improve

local diameter predictions due to the joint regression of common structural features (in this study: spread, qmd, diameter 10% quantile). In contrast, the direct fits used solely the data of single sample plots and thus performed less accurately if the sample did not capture the stand's diameter distribution well, e.g., due to a low number of sampled trees. This is also confirmed by the low improvement of the simultaneous PPM for the local pole stage stands, whose area was generally lowest, and by the higher performance of the simultaneous PPM based on data from sample plots with fewer than 17 (weighted Weibull) and 19 trees (truncated Weibull). Our findings are further supported by Gove (2003) and García (2006), who showed that a higher number of trees reduced the bias of estimated stand-level diameter distributions. We thus conclude that the simultaneous PPM can be recommended to improve diameter predictions of local stands from large-scale sample data, especially for sample plots that contain little stand-level information.

Regarding the prediction of tree species composition at the local scale, the errors of the Random Forest were fairly balanced across the tree species groups. In contrast, the weighted sample data led to a stronger over- than underestimation of species occurrence at the stand level. This can be explained by a) the spatial aggregation of tree species within the stands, b) the species mixture within the stands, and c) an insufficient number of sample replicates to average out over- and underestimation: The spatial aggregation of individual tree species may not be random (or uniform) but clustered in near-natural stands (Szwagrzyk and Czerwczak, 1993). Such clusters may lead to over- or underestimations in the weighted sample data since clustered tree species are likely to be sampled less frequently than randomly or uniformly distributed species. Thus, a large difference between over- and underestimation of the occurrence of a tree species can indicate a clustered distribution of that species, and highly diverse species mixtures may reinforce this effect. For example, spruce stands are often monospecific, resulting in smaller differences between over- and underestimation, while (managed) beech stands are more diversely mixed, resulting in larger differences between over- and underestimation. It is noteworthy that the species composition estimated from NFI sample data can be considered as unbiased (Fuller, 2009): following the Law of Large Numbers (Dekking et al., 2005), a large number of sample replicates should balance over- and underestimation, even if tree species are highly clustered within the stands. However, since we only used 100 sample replicates, this effect may be not visible in the results of the weighted sample data. In contrast, the Random Forest balanced over- and underestimation for each sample set due to bagging (bootstrapping and aggregation), resulting in more stable and thus accurate predictions, which underlines its suitability for predicting stand species composition using data from small sample plots.

4.3. Modelling approaches

4.3.1. Diameter distributions

We investigated two parameter prediction methods (PPMs) and showed that the simultaneous (GLM-like) PPM (Cao, 2004) is superior compared to the traditional approach. While Cao (2004) and Rätty et al. (2021) found a similar result, these studies evaluated diameter predictions based on sample (and airborne laser scanning) data only. We used field data of completely measured stands to evaluate the diameter predictions derived from sample data and demonstrated the advantages (but also disadvantages) of the simultaneous PPM compared to simpler approaches. Below, we discuss the methodological background and underlying model assumptions.

The PPM approaches have the following advantages: (a) they link diameter information from multiple sample plots in regression models, (b) they can be flexibly used for diameter predictions inside and outside the range covered by the data used for model building (Rätty et al., 2021) and (c) they can be used for local (based on each sample plot) and generalised (i.e., stratum) scale predictions. Further, a small number of sample plots may be sufficient for model building if they cover a

representative gradient of the potentially possible distributions. In addition, our analysis shows that the simultaneous PPM approach outperforms stand-level predictions compared to simpler approaches based on data from small sample plots in most cases. However, we did not compare the PPM to other statistical methods that may perform equally well or even better. For example, a closely related approach to the PPM is the parameter recovery method (PRM), in which stand variables (e.g., moments, percentiles) are used to analytically recover the parameters of a PDF (Hyink and Moser, 1983). Siipilehto and Mehtätalo (2013) showed that PRM outperformed PPM for Scots pine stands in Finland, while Chen et al. (2019) found advantages of the PPM compared to the PRM for fir plantations in China. In addition, entirely non-parametric approaches are possible. For example, Rätty et al. (2021) used nearest neighbouring to predict diameter distributions based on NFI and airborne laser scanning data and showed that this approach can outperform the simultaneous PPM. Magnussen and Renaud (2016) used multidimensional scaling for stand-level (local) and stratum-level (generalised) predictions of diameter distributions. However, in contrast to the PPM and PRM such non-parametric approaches must be based on even larger datasets and are limited to the data used for model fitting.

Our PPM approaches are based on a number of assumptions:

- (a) The existence of a probability density function (PDF) that describes the diameter distributions reasonably well. The choice of the PDF is left to the researcher and usually has no fundamental biological justification (Siipilehto and Mehtätalo, 2013). We chose the Weibull distribution since it has been proven to be flexible and efficient in describing unimodal diameter distributions (de Lima et al., 2015; Palahí et al., 2007). However, we also showed that the choice of the Weibull function was associated with an Error index of around 8–10% compared to the representation of the diameter distributions of stands by non-parametric KDE (cf. Fig. 3). Other PDFs such as gamma, beta, lognormal, or Johnson's SB have been considered to be suitable for describing tree diameter distributions (Hafley and Schreuder, 1977; Maltamo et al., 1995; Palahí et al., 2007), and in specific situations they may be more suitable than the Weibull PDF. Magnussen (1986) found that the Weibull distribution is less suited to describe diameter distributions in unthinned and heavily thinned stands. Hafley and Schreuder (1977) recommended Johnson's SB due to its flexibility in terms of skewness and kurtosis, and also de Lima et al. (2015) showed that negatively skewed as well as heavy-tailed diameter distributions were better described by the logit-logistic distribution (proposed by Wang and Rennolls, 2005) than by the Weibull PDF. Our results confirmed these considerations because negatively skewed stands were often associated with large errors for the Weibull-based PPM approaches (cf. Fig. S5).
- (b) We assumed that the PDF parameters vary linearly with stand characteristics and did not test non-parametric predictors (e.g., smoothing splines) since the flexibility of such non-parametric methods is usually accompanied by a loss of interpretability and may lead to overfitting. Also, Rätty et al. (2021) found that including smoothing terms did not improve the prediction performance of the simultaneous PPM (termed GLM in their paper).
- (c) The simultaneous PPM assumes independency of the trees within a sample plot. However, spatially related trees are linked, e.g., through competition, facilitation and site-specific conditions. Including random effects in model formulation can take such relationships into account (cf. Rätty et al., 2021), but they also limit predictions to the local scale (sample plots of the training data) and make parameter optimisation more difficult.
- (d) We restricted our analysis to unimodal diameter distributions and excluded the commonly observed bi- and multi-modal ones. Such bimodal diameter distributions must be described by more

flexible methods than the unimodal Weibull PDF (e.g., mixed distributions, cf. Mey et al., 2021). However, bimodal diameter distributions were less frequently observed in the data (also because only trees ≥ 8 cm (EFM) or ≥ 4 cm (NFI) were taken into account), and mixed distributions are more complex and more difficult to estimate.

- (e) We restricted our PPM approaches to the two best shape and scale predictors, as this allowed for consistent parameter optimisation. An even more exhaustive analysis considering other moments and percentiles of the diameter samples and using more than two explanatory variables could lead to even better results.

4.3.2. Tree species composition

The Random Forest approach generally outperformed the multinomial logistic regression and the direct use of the sampled species composition, which corresponds to the findings by Mey et al. (2022). It performed considerably better with four explanatory variables compared to the multinomial logistic regression with ten variables. This indicates a nonlinear relationship between the predictors and the response, and the Random Forest appears advantageous since classification models with fewer variables are generally easier to interpret (Evans et al., 2011). Due to its easy implementation, fast computing time and (often) good prediction performance, Random Forests have been gaining attention in forestry in recent years (Evans et al., 2011; Prasad et al., 2006). However, the approach requires comprehensive, large data sets, and thus it is often applied to data-rich sources such as LiDAR data or satellite imagery (e.g., Hologa et al., 2021; Plakman et al., 2020). Only a few studies have applied it to large-scale (national) forest inventory data (Costanza et al., 2018; Jevšenak and Skudnik, 2021; Kilham et al., 2018). For example, Kilham et al. (2018) demonstrated that a Random Forest based on inventory data improved the predictions of harvesting probabilities of sample trees in Southwest Germany. To the best of our knowledge, we showed for the first time that a Random Forest based on NFI data outperformed other methods to predict stand species composition at the local and generalised (i.e., forest type-specific) scale.

Yet, the Random Forest approach is limited in several aspects: (a) effect directions of the Random Forest predictions may be unstable, i.e. they can change rapidly depending on the data used for model fitting, and therefore the interpretability of the effects of the single explanatory variables is low compared to parametric methods, e.g., multinomial logistic regression; (b) it cannot be used for extrapolation beyond the range covered by the data used for model fitting, and (c) the standard Random Forest approach does not account for dependencies in the data. Our final Random Forest model included the sample plot ID to account for stand-specific effects (e.g., reproduction advantages of the dominant species) and therefore directly accounted for the dependency of trees nested at the same sample plots, but this limited the predictions to the investigated stands (in-sample). Other approaches that include random effects are probably more suited to account for the nested data structure of NFIs, e.g., a generalised mixed-effects Random Forest as recently proposed by Pellagatti et al. (2021) or a mixed-effects multinomial logistic regression approach (Hedeker, 2003). However, for our study these limitations seemed acceptable as we focused on stand-level predictions for data from each sample plot (local) or joint sample plots (generalised). We suggest that for comprehensive data sets covering a large explanatory gradient (which is usually the case with NFI data), a Random Forest is quite promising to provide good predictions of stand species composition.

4.4. Initialisation of dynamic forest models

Climate change affects forest dynamics across multiple scales (from small forest patches to stands to landscapes to regional and national scales), and thus integrative dynamic forest models including small- and large-scale processes are needed (e.g., Coates et al., 2003; Seidl et al.,

2012). However, at larger spatial scales such as countries, detailed initial data for models that account for small-scale processes (i.e., growth and mortality of individual trees at specific locations) are usually not available. We demonstrated that the (adjusted) methodology by Mey et al. (2021) is suitable for using NFI sample data for stand-level model initialisation at large spatial scales (e.g., countries, ecoregions), and we quantified the errors that originate from this type of initialisation. The method by Mey et al. (2021) allows to generate (stand-level) tree lists (diameter and species) at diverse spatial scales and thus should be applicable to most individual-tree based to stand-level forest models (e.g., Bugmann, 1996; Coates et al., 2003; Pretzsch et al., 2002; Seidl et al., 2012). Depending on the modelling objective, such stands can be generated at the local or generalised scale (e.g., development stage, forest type).

Generalised stand descriptions can be used to investigate the effects of regional forest management and climate change at the level of a forest stratum, e.g., even-aged beech-dominated forests (Huber et al., 2021; Mey et al., 2022). Analysing general trends of management and climate effects may be particularly useful for federal policy making and federal management plans (FOEN, 2021). In contrast, local stand descriptions can be used to investigate stand-specific effects of management and climate impacts, and ultimately give one-to-one recommendations how these stands can be managed. For example, Thrippleton et al. (2021) used local stand descriptions in a simulation study to evaluate future ecosystem service provision under different management intensities and climate change to derive decision support in forest enterprises in Switzerland. Local and generalised stand descriptions based on NFI data can cover a wide range of site and legacy effects, which makes them useful for the initialisation of process-based and empirical forest models at large scales.

For model initialisations at the local scale, however, it should be taken into account that the errors can be considerably higher than at the generalised scale. For example, to make the method useful for the initialisation of spatially explicit forest landscape models with a gridded landscape representation (e.g., Petter et al., 2020), a spatial tree pattern must be added, e.g., derived from LiDAR data (Wang et al., 2020; Wiggins et al., 2019); alternatively, local stand descriptions (e.g., on the 1.4 km $\times$ 1.4 km-NFI grid in Switzerland) can be imputed across the landscape. To this end, landscape builder tools can be used to stochastically assign representative local stand descriptions to each cell of a landscape model (cf. Dijak, 2013).

Overall, our initialisation approach can improve the accuracy and precision of model simulations since the simulation of stand dynamics at the decadal time scale critically depends on the initialisation data (Pretzsch, 2009; Temperli et al., 2013). At smaller scales (e.g., single tree level), the representation of physiological processes is likely to be even more sensitive to the initialisation (Poschenrieder et al., 2013). Consequently, simulating growth and mortality of single trees across larger scales may accumulate uncertainty/bias of initialisation (Fontes et al., 2011). Due to limited empirical data for model initialisation, simulation approaches at larger spatial scales often use a “spin-up” to generate the initial state from the model itself (Elkin et al., 2013; Thom et al., 2018). However, such simulated initial conditions often differ from the observed vegetation and do not include legacy effects (Thom et al., 2018). In contrast, the approach presented here accurately depicts the current state of forests using large-scale empirical data, and thereby minimises potential initialisation biases.

4.5. Conclusions

This study is the first to evaluate PPM models based on large-scale NFI samples and their accuracy in predicting stand-level data at the local and generalised scale. We showed that the simultaneous PPM and the Random Forest generally outperformed simpler approaches, although simpler models may be preferable in certain situations (e.g., pooling for generalised uneven-aged stands). We evaluated model

performance using sample data from 271 stands; applying our methods to (NFI) data sets with several thousand sample plots or to data sets with higher forest diversity (e.g., NFI data from the tropics with higher structural and species diversity) as part of a future study would likely improve the performance of the simultaneous PPM and the Random Forest even further. Prediction errors were considerably higher at the local than at the generalised scale, which should be considered when using the models for local scale applications (e.g., predicting single stand descriptions in forest enterprises). Overall, we provide a basic methodology to derive stand descriptions using data from small sample plots that is easily transferable to other large-scale data sets in other regions (e.g., by adjusting the calliper thresholds according to the sampling design of other NFIs), and give instructive guidance for model building that help to avoid methodological pitfalls using the approach of Mey et al. (2021).

Our results indicate that the (adjusted) approach by Mey et al. (2021) is an important step towards a more targeted and detailed initialisation of dynamic forest models at multiple spatial scales. Such cross-scale model applications may help to involve stakeholders working at different administrative levels and geographic scales, and foster the communication of scientific results for management guidance. In the face of changing future boundary conditions, the joint application of modelling approaches that operate on diverse spatial and hierarchical scales is needed (e.g., Boulanger et al., 2022; Irauschek et al., 2021). Our multi-scale initialisation approach may help to standardise and improve such joint model initialisations across spatial scales, and thus ultimately increase precision and trust in model predictions.

CRedit authorship contribution statement

Reinhard Mey: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing – original draft. **Christian Temperli:** Supervision, Conceptualization, Writing – review & editing. **Jonas Stillhard:** Conceptualization, Data curation, Writing – review & editing. **Jens Nitzsche:** Data curation. **Esther Thüri:** Project administration, Funding acquisition. **Harald Bugmann:** Supervision, Writing – review & editing. **Jürgen Zell:** Supervision, Conceptualization, Methodology, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foreco.2023.121155>.

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